Disciplinary Astronomy

University of Groningen

Your interests: **Space exploration and astronomy**

Reference programme: Astronomy, University of Groningen ↗

Astronomy and astrophysics

- Introduction Astronomy
- Physics of Stars
- Physics of Galaxies
- Radiative Processes in Astrophysics
- Advanced Electrodynamics
- Astrophysical Hydrodynamics
- Interstellar Medium

Physics and mathematics

- Mechanics and Relativity — 10 EC
- Electricity and Magnetism — 10 EC
- Thermal Physics — 10 EC
- Quantum Physics 1
- Quantum Physics 2
- Waves and Optics
- Calculus 1 (for Physics)
- Calculus 2 (for Physics)
- Linear Algebra (for Physics)
- Mathematical Physics
- Complex Analysis

Computing, data and observation

- Introduction to Programming and Computational Methods
- Numerical Methods
- Statistics for Astronomy
- Observational Astronomy

Exploration, instruments and responsible technology

- Minor: Instrumentation & Informatics — 30 EC

Earth, planets and life

- Introduction to Planetary Science

Research and academic development

- Physics Lab: Skills
- Astronomy Bachelor Research Project — 15 EC

Selected minor route: Instrumentation & Informatics (30 EC).

UCR — Mathematics, Data & Observation

greatest disciplinary depth

Your interests: **Space exploration and astronomy**

Physics and mathematics

- Advanced Calculus
- Linear Algebra
- Mathematical Methods
- Logic & Discrete Mathematics
- Linear Systems

Computing, data and observation

- Introduction to Programming
- Computational & Numerical Methods
- Probability & Statistics
- Earth Observation
- Introduction to Data Science
- Machine Learning
- Image Processing & Computer Vision
- Neural Networks & Deep Learning
- Optimization Methods in AI
- Reinforcement Learning
- Algorithms & Data Structures
- Networks & Operating Systems
- Database Management Systems
- Earth System Science Tools

Exploration, instruments and responsible technology

- Robotics
- Consumer Product Design

Earth, planets and life

- Sustainability Essentials I
- Environmental Biogeochemistry

Research and academic development

- Personal & Professional Development

UCR does not offer astronomy or physics courses; this pathway builds the closest feasible depth through mathematics, computing, data and observation.

P-002 | Fall 2026 cohort

UCR — Space, Earth & Technology

more balanced across your interests

Your interests: **Space exploration and astronomy**

Physics and mathematics

- Linear Algebra
- Mathematical Methods
- Logic & Discrete Mathematics

Computing, data and observation

- Introduction to Programming
- Computational & Numerical Methods
- Probability & Statistics
- Earth Observation
- Introduction to Data Science
- Machine Learning
- Image Processing & Computer Vision
- Optimization Methods in AI
- Earth System Science Tools

Exploration, instruments and responsible technology

- Robotics
- Consumer Product Design
- Renewable System Design
- AI Ethics

Earth, planets and life

- Sustainability Essentials I
- Sustainability Essentials II
- Dynamic Earth
- Environmental Biogeochemistry
- Earth's Resources
- Climate Change

Research and academic development

- Research in a Sustainable Delta
- Personal & Professional Development

UCR does not offer astronomy or physics courses; this pathway combines quantitative tools with Earth science and exploration technology.

P-002 | Fall 2026 cohort

UCR — Worlds, Life & Exploration

broader thematic pathway

Your interests: **Space exploration and astronomy**

Physics and mathematics

- Mathematical Methods

Computing, data and observation

- Introduction to Programming
- Earth Observation
- Introduction to Data Science
- Machine Learning
- Image Processing & Computer Vision
- Neural Networks & Deep Learning
- Earth System Science Tools

Exploration, instruments and responsible technology

- Robotics
- Consumer Product Design
- Renewable System Design
- AI Ethics

Earth, planets and life

- Sustainability Essentials I
- Sustainability Essentials II
- Dynamic Earth
- Environmental Biogeochemistry
- Earth's Resources
- Climate Change
- Introduction to Biodiversity
- The Science of Evolution
- Foundation of Life, Health & Biomedicine
- Biochemistry

Research and academic development

- Research in a Sustainable Delta
- Personal & Professional Development

UCR does not offer astronomy or physics courses; this pathway broadens into Earth systems, life science, data and exploration technology.